Fiscal Year:	FY 2023	**Task Last Updated:**	FY 05/16/2023
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Project Title:	Time and Space, Determining the Interactions Between the Circadian Clock and Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	01/04/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/04/2023 per NSSC information (Ed., 4/6/23) NOTE: End date changed to 7/25/2022 per NSSC information (Ed., 8/9/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC18K1466		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The circadian clock, an internal timekeeping mechanism, enables organisms to temporally organize their molecular and biochemical activities so that they are optimally timed with environmental conditions. The circadian clock contributes to overall fitness in plants and animals. Our preliminary results support a connection between plant gravity sensing and responses and the circadian clock, which has not been explored/reported previously. We propose to examine this interaction in both directions in Arabidopsis plants: investigating the effects of the circadian clock on gravity perception and responses and the system-level effects of simulated microgravity on the molecular clock. In early spaceflight studies, the impacts of microgravity on the circadian clock and the subsequent effects on physiology in humans and other organisms were recognized, as well as the potential for profound implications for crew performance. In mammalian studies, there is evidence that microgravity and recovery from microgravity affect core components of the mammalian circadian clock. Yet a full examination of the molecular interplay between gravity and the circadian clock has not been reported in any organism. As with crew-performance, a disrupted circadian clock has a negative impact on plant growth and fitness. In plants, almost all aspects of growth, physiology, and environmental responses are modulated by the circadian clock. Therefore, understanding the interplay between changes in gravity, the circadian clock, and circadian-regulated output will have important implications for plant growth in future space missions. The experiments outlined here will identify not only the effects of changes in gravity levels on the core clock components but will also examine the genome-wide downstream effects on transcriptional timing. We observed that the time of day a gravitropic stimulus is provided affects the magnitude of the response. Furthermore, this response is altered in a circadian mutant. Based on these findings we hypothesize that, like other environmental stimuli, the response to gravity is gated by the circadian clock. We also examined transcriptomic datasets in GeneLab and found that core clock components are differentially expressed in response to gravity in multiple studies in Arabidopsis. This suggests that the circadian clock may be sensitive to changes in gravity. To examine how the interaction between the circadian clock and gravity is affected under simulated microgravity we will utilize the random positioning machine to evaluate the following hypotheses. Hypothesis 1: Gravity sensor(s) is/are controlled by the circadian clock. Therefore, the time of day the plants are transferred to microgravity will affect their response and the response to gravity will be altered in plants with mutations in the circadian clock. Objective: Determine the effects of the time of day and the circadian clock on gravitropic responses. Hypothesis 2: Gravity signals modulate the Arabidopsis circadian clock and circadian-regulated outputs. Therefore, clock components and/or clock outputs will be altered under simulated microgravity. Objective: Investigate how simulated microgravity affects the circadian clock and circadian-regulated outputs. This proposal focuses on the interaction between gravity and the circadian clock in a plant system, yet many principles of circadian circuits span across multiple organisms. Transcriptomic data will be generated and deposited into the GeneLab database. The system-level understanding of gravity perception and responses and the intersection with the circadian clock acquired here will contribute towards adapting plants to spaceflight and may also improve our understanding of the impacts of space travel on other organisms. Training of new generation US Space Biologists will focus on data analysis and integration of data sets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has two investigation tracks which have an impact on life on Earth. The first aspect is to understand how the time of day and the circadian clock affect the response to gravity in plants. The circadian clock controls many aspects of growth and development in plants and does this through daily regulation of transcription, translation, and other biochemical activities. Since most of the molecular components of the plant show a daily rhythm in their activities, the pathways and responses that depend on them also vary in efficiency throughout the day. We are using this underlying variation in biochemistry and physiology to uncover the pathways that plants use to sense and respond to changes in gravity. Understanding plant responses to gravity can have significant impacts on life on Earth as plant root growth, controlled by gravity, is an important factor in the ability of plants to access water. Our second investigation track focuses on the inverse connection--does gravity affect the circadian clock? The circadian clock of plants and animals enables organisms to optimize their response to the environment coordinating their internal molecular activities with externally recurring events such as sunrise and sunset or the changing of seasons. The internal circadian clock has been shown to enhance fitness in many species, and an altered clock can affect performance (e.g., yield in plants, response time in humans, and resistance to disease in both plants and animals). This molecular clock has developed in organisms within the confines of Earth with an ~24 h diel cycle and under ~1g of gravity. We are investigating if changes in gravity affect the functioning of the molecular clock, either through altering the oscillator itself or the output regulated by the circadian oscillator. If the clock is impacted by gravity, this can be used to understand the clock mechanisms. Insights into how the clock works and how it is connected to the downstream activities it regulates can be used to enhance the performance of crops, particularly as changing climate conditions allow us to grow crops in new environments. Two critical changes in plant growth are the poleward movement of cultivated areas and the expansion of growing seasons throughout the year. Both of these aspects involve components that are regulated by the circadian clock. Therefore, understanding how the clock functions can provide us with strategies for enhancing plant adaptability to new environmental conditions.
	The major goals of this project are to answer two fundamental questions: Aim 1 -- Does the circadian clock affect the response to changes in gravity? Aim 2 -- Does gravity affect the circadian clock? AIM 1: We completed the evaluation for Aim 1 (of the role of the circadian clock and time of day in the response of Arabidopsis plants to gravity). Importantly, in collaboration with NASA Kennedy Space Center (KSC) personnel, we established a method for growing plants in unique lighting conditions while on the Random Positioning Machine (RPM). We optimized this setup and can now efficiently evaluate the response of plants to microgravity when the gravity stimulus is changed at different times of day. Finally, we have compared the response to gravity in plants with defective circadian clocks and have observed the effects of the response in these plants. The experiments related to Aim 1 (Does the circadian clock affect the response to changes in gravity?) were completed and resulted in two published works: (1) A methods book chapter on how to appropriately account for time in

Task Progress:	microgravity experiments, and (2) A research journal article on the effects of time of day and the circadian clock on gravitropic responses. Briefly, the time of day and the circadian clock affect the gravitropic response. Therefore, the time of day experiments are performed is a critical feature that should be considered for gravitropic experiments. Additionally, we showed that it is likely not the statolith changes that drive this change in response. In this reporting period, we completed all experiments related to question 1 (Aim 1). These findings identify additional questions (which will not be answered by this proposal), including if changes in auxin levels or perception drive the changes in responses we observe. Additionally, we identified a major technical hurdle that needs to be resolved. Measurements of roots on agar plates in the RPM are limited by the two-dimensional approach that is commonly used -- seeds grown on the top of the media and images that are in two dimensions only. This approach could not capture the many roots we observed to grow into or out of the media when placed on the RPM. Identifying a method to measure the three-dimensional growth of roots on the RPM would improve our ability to identify how plants respond to microgravity in a more quantitative manner. AIM 2: In answering Aim 2: how simulated microgravity affects the circadian clock, we have completed a 48-hour time course of plants collected every 2h in constant light from control and simulated microgravity conditions on the RPM. This data provides us with two opportunities that we are preparing for publication: (1) the effects of spike-in normalization on analysis of circadian rhythms from RNA-seq data, and (2) the effects of simulated microgravity on the circadian clock. The effects of spike-in normalization on analysis of circadian rhythms from RNA-seq data: A fundamental assumption in the analysis of RNA-seq data is that the expression of most genes does not change. That is, the fundamental rate of transcription, the activity of RNA Pol II is consistent between the conditions or time points being analyzed. However, we know that for many environmental stresses, genetic disruptions, and even different times of day in the same species transcription levels themselves change. In mouse cells, transcription is most active just after dusk, and changes in chromatin coordinate this tight temporal restriction. Evidence for this coordinated transcription in plants has not been shown. However, there are several studies with evidence that supports the possibility that transcriptional levels vary throughout the day. We have observed this bias in our own samples when comparing plants grown in the early morning when photosynthesis is highly active (and is at least 30% of the transcriptome) to samples grown at night. If transcriptional levels are substantially different, this invalidates many of the commonly used mechanisms for normalizing RNA-Seq data as they assume that the total transcriptional levels are not changing. Fortunately, an easy fix for this can be done by adding spike-ins of known RNA amounts to set amounts of tissue before RNA extraction. We have done this with the control and RPM-based samples from this experiment. We can analyze the data both with, and without, normalizing to the spike-ins. This analysis shows that there is an underlying rhythm to the gene expression that is apparent when normalizing to the spike-ins but is not present in the traditional normalization. This rhythm shows two peaks of expression. One at just before dusk and one just before dawn. Further data analysis, compared with the two different normalization methods, reveals significant differences in the resulting analysis. We observed that substantially more genes are identified as cycling, using Jonckheere-Terpstra-Kendall (JTK) analysis, with the traditionally normalized data than with the spike-ins. The effects of simulated microgravity on the circadian clock: A similar analysis has been performed for RPM samples. Here we are comparing the control to the treatment and asking if any genes are different in response to growth on the RPM. We have done this with traditional normalization and our spike-in data set. Both identify different gene sets. We find several candidate genes, some of which are central circadian components disrupted under simulated microgravity. Moreover, we observe genes with a phase advance and arrhythmic, both of which are consistent with disruptions in part of the circadian clock. We observe arrhythmic genes. These are of particular interest as they may be directly connected to circadian components which are affected under simulated microgravity.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Articles in Peer-reviewed Journals	Fountain LL, Gilliham M, Amitrano C, Arouna N, Barker RJ, Böhmer M, Braun M, Brereton NJB, Brocato RL, Bunchek JM, Canaday ELJ, Caplin N, Castaño P, Chamberlain C, Decourteix M, Del Bianco M, De Micco V, Doherty CJ, Franke MF, Fuentes S, Gilroy S, Harrison L, Hasenstein KH, Hauslage J, Herranz R, Iyer-Pascuzzi A, Izuma DS, Junya K, Kiss JZ, Legué V, Lloyd JPB, Maffei ME, Massa GD, Meyers AD, Perera IY, Poulet L, Roychoudry S, Sena G, Shippen DE, Stoochnoff J, Takahashi H, Wyatt SE, Blancaflor EB. "Expanding frontiers: Harnessing plant biology for space exploration and planetary sustainability." New Phytol. 2026 Jan;249(2):656-69. https://doi.org/10.1111/nph.70662 ; PMID: 41295882 , Jan-2026